

MSCIT SEM 1 - MONGODB

Chapter-3

Session Management

- Understanding HTTP sessions.
- Understanding PHP native session handling.
- Implementing session handling with MongoDB.
- Putting Session Manager.
- Building user authentication module, creating login, logout and user profile.

*** Explain HTTP sessions.**

- HTTP session is a sequence of network request-response transaction.
- An http client request by establishing a transmission control port on a server.
- Http server listen client requests from that port and after receive the request server send back some response message.
- The http support transport layer and the url of a page contains the domain name and the port number.
- An http is stateless protocol and it does not require http server about each user for the duration of multiple request.

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- Some web applications implements states or server side session using web forms, http cookies or http hidden variables.
- HTTP sessions is an industry standard feature that allows Web servers to maintain user identity and to store user-specific data during multiple request/response interactions between a client application and a Web application.
- HTTP sessions preserves:
 - Information about the session itself (session identifier, creation time, time last accessed, etc.)
 - Contextual information about the user (client login state, for example, plus whatever else the Web application needs to save)
- HTTP sessions are useful for load balancing.

*** Explain PHP session handling.**

- In PHP session handler a special class is used for session handling.
- There are 7 method which are used for internal session handling like open, close, read, write, destroyed, GC and create_sid.
- Session.save-handler class will wrap the internal save handler and configure the directive which files by default.
- When a plain instance(object) of session handler is set as a save handler using session_set_save_handler() then it will wrap the current save handlers.

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Methods: -

1. Session handler_create_session_id:- it return a new session Id
2. Session handler_open():- it initialize a session .
3. Session handler_close():-it close the session.
4. Session handler_read():- it read the session data.
5. Session handler_write():- it write the session data.
6. Session handler_gc():- it clean up old session.
7. Session handler_destroyed:- it destroyed a session.

*** Understanding PHP native session handling:**

Php handle on session natively hear it configure the session on server (php.ini file) and not in the application.

So, if we have several websites on one server then we only need to configure the server for the changes to be applied on all the websites.

There are different libraries can be used for native session handling, hear you can also create a new library that starts a native php session and allows to store and retrieve different things from it.

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With the release of PHP4, session management was introduced as an extension to the PHP language. PHP provides several session-related functions, and developing applications that use PHP sessions is straightforward.

The three important features of session management are mostly taken care of by the PHP scripting engine.

How to use PHP sessions? How sessions are started and ended? and How session variables are used? We list the PHP functions for building session-based web applications. Because not all browsers support cookies, and some users actively disable them.

When a user first enters the session-based application by making a request to a page that starts a session, PHP generates a session ID and creates a file that stores the session-related variables.

PHP sets a cookie to hold the session ID in the response the script generates. The browser then records the cookie and includes it in subsequent requests.

Between the browser and the server when initial requests are made to a session-based application the out-of-the-box configuration of PHP session management uses disk-based files to store session variables.

Using files as the session store is enough for most applications in which the numbers of concurrent sessions are limited.

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Starting a Session PHP provides a `session_start()` function that creates a new session and subsequently identifies and establishes an existing one. The first time a PHP script calls `session_start()`, a session identifier is generated, and, by default, a Set-Cookie header field is included in the response

The response sets up a session cookie in the browser with the name `PHPSESSID` and the value of the session identifier.

The PHP session management automatically includes the cookie without the need to call to the `setcookie()` or `header()` functions.

The session identifier (ID) is a random string of 32 hexadecimal digits, such as `fcc17f071bca9bf7f85ca281094390b4`. As with other cookies, the value of the session ID is made available to PHP scripts in the `$HTTP_COOKIE_VARS` associative array and in the `$PHPSESSID` variable.

When a new session is started, PHP creates a session file. With the default configuration, session files are written in the `/tmp` directory using the session identifier, prefixed with `sess_`, for the filename.

However, if the identified session file can't be found, `session_start()` creates an empty session file.

Session Variables need to be registered with the `session_register()` function that's used in a session. If a session has not been initialized, the `session_register()` function calls `session_start()` to open the session file.

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Once registered, session variables are made persistent and are available to scripts that initialize the session. PHP tracks the values of session variables and saves their values to the session file: there is no need to explicitly save a session variable before a script end.

*** Implementing Session handling with MongoDB**

To implement session with MongoDB you have to use connect framework and express framework.

First of all, you have to connect-mongodb.

API

MongoDBStore

This module exports a single function which takes an instance of connect (or Express) and returns a MongoDBStore class that can be used to store sessions in MongoDB.

The MongoDBStore class has 2 required options:

- (1) uri: a MongoDB connection string
- (2) collection: the MongoDB collection to store sessions in.

If you pass in an instance of the [express-session module](#) the MongoDBStore class will enable you to store your Express sessions in MongoDB.

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In many circumstances, connect-mongo will not be the only part of your application which need a connection to a MongoDB database. It could be interesting to re-use an existing connection.

Alternatively, you can configure connect-mongo to establish a new connection.

A Session is a way to store the information that Is used across multiple pages not stored in user Computer.

The web server does not know who you are or what you do because the http is having stateless protocol and so does not maintain the state

Session is used to store the information in the format of session variables.

Here user can store the information that is used for multiple pages like username, back color, etc...

The session is available by default until the user closes the browser.

If you want to store information permanent then you can store it in database.

In php session_start () function is used to start a session.

You have to write this function very first thing in your document.

If you want to store session variable value then a globle variable \$_session is used.

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If you want to change the session variable value then you have to just override it.

If you want to remove all global session variable & destroyed the session then you have to use session_unset() & session_destroy().

Ex1: -

Website Counter

```
<?php
Session_start();
If(isset($_SESSION['counter']))
{
$_SESSION['counter']+=1;
}
Else
{
$_SESSION['counter']=1;
}
$msg="you have visited this page".$_SESSION['counter'];
$msg="in this session";
?>

<html>
<head>
<title> counter</title>
</head>
<body>
<?php echo ($msg); ?>
</body>
</html>
```


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Ex-2:-

```
<html>
```

```
<body>
```

```
<?php
```

```
//remove all session variable
```

```
Session_unset();
```

```
//destroyed the session
```

```
Session_destroy();
```

```
?>
```

```
</body>
```

```
</html>
```